Anna **Becchi**

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Research Interests

I work on program verification with Constrained Horn Clauses (CHCs): my recent works address certified Horn solving, and improving convergence of solving algorithms such as Spacer (PDR), and Transition Power Abstraction. During my PhD at FBK and the University of Trento, I developed abstraction techniques for the reverse engineering of timed and hybrid systems, applied to railway interlocking systems.

Keywords Constrained Horn Clauses, Certification, Lean, Model-Based Projection, Quantifier Elimination, Polyhedral Abstraction

Experience

Università della Svizzera italiana (USI) — Lugano, Switzerland

Postdoctoral Fellow — Oct 2025 – present

- Formal Verification and Security Lab, led by Prof. Natasha Sharygina.
- Certified Horn solving and CHC-based design, model-based projection, and convergence of CHC solving algorithms.

Fondazione Bruno Kessler (FBK) — Trento, Italy

Research Assistant — Nov 2024 – Sep 2025

- Formal Methods Unit, led by Stefano Tonetta.

Fondazione Bruno Kessler (FBK) — Trento, Italy

Software Developer — Sep 2019 – Oct 2021

- Embedded Systems Unit (former), led by Alessandro Cimatti.

Education

Fondazione Bruno Kessler & University of Trento — Trento, Italy

PhD in Computer Science, with distinction — Nov 2021 – Sep 2025

- Thesis: *Abstractions for Reverse Engineering — Validating the Computerization of Relay-Based Railway Interlocking Systems*. Supervisor: Alessandro Cimatti.
  I developed *Abstraction Modulo Stability* (AMS), a framework for the extraction of properties and reverse engineering of infinite-state timed and hybrid systems: I applied AMS on a project with the Italian Railway Network to validate the migration from analog relay-based systems to software-based interlockings.

University of Udine — Udine, Italy

MSc in Computer Science, 110/110 with honors — Sep 2017 – Jul 2020

- Thesis: *P-Stable Abstractions for Hybrid Systems*. Supervisor: Prof. Angelo Montanari.

University of Parma — Parma, Italy

BSc in Computer Science, 110/110 with honors — Sep 2014 – Sep 2017

- Thesis: *NNC Polyhedra: a New Representation and Conversion Algorithm*. Supervisor: Prof. Enea Zaffanella.

Research Visits

University of Waterloo — Waterloo, Canada

International Graduate Visiting Student — May – Aug 2024

- Hosted by Prof. Arie Gurfinkel. CHC solving and model-based projection for Linear Rational Arithmetic.

Awards

Radhia Cousot Young Researcher Best Paper Award — Static Analysis Symposium (SAS 2019)
For the paper *Revisiting Polyhedral Analysis for Hybrid Systems*

Publications

Also on dblp.

JOURNAL ARTICLES

Abstraction Modulo Stability — FMSD 2024
A. Becchi, A. Cimatti

P-Stable Abstractions of Hybrid Systems — SoSyM 2024
A. Becchi, A. Cimatti, E. Zaffanella

PPLite: Zero-Overhead Encoding of NNC Polyhedra — Inf. & Comp. 2020
A. Becchi, E. Zaffanella

CONFERENCE PAPERS

PyCHC: a Framework for Certified Horn Solving and CHC-based Design — CAV 2026
A. Becchi, M. Blicha, R. Otoni, N. Sharygina

Syntactically Convex Model-Based Projection for Linear Real Arithmetic — TACAS 2026
A. Becchi, G. Fedyukovich, A. Gurfinkel, L. Nachmanson

Testing the Migration from Analog to Software-Based Railway Interlocking Systems — CAV 2024
A. Becchi, A. Cimatti, G. Scaglione

Searching for i-Good Lemmas to Accelerate Safety Model Checking — CAV 2023
Y. Xia, A. Becchi, A. Cimatti, A. Griggio, J. Li, G. Pu

SMT-Based Stability Verification of an Industrial Switched PI Control Systems — DSN-W 2023
S. Basagiannis, L. Battista, A. Becchi, A. Cimatti, G. Giantamidis, S. Mover, A. Tacchella, S. Tonetta, V. A. Tsachouridis

Abstraction Modulo Stability for Reverse Engineering — CAV 2022
A. Becchi, A. Cimatti

NORMA: a Tool for the Analysis of Relay-Based Railway Interlocking Systems — TACAS 2022
A. Amendola, A. Becchi, R. Cavada, A. Cimatti, A. Ferrando, L. Pilati, G. Scaglione, A. Tacchella, M. Zamboni

A Model-Based Approach to the Design, Verification and Deployment of Railway Interlocking System — ISoLa 2020
A. Amendola, A. Becchi, R. Cavada, A. Cimatti, A. Griggio, G. Scaglione, A. Susi, A. Tacchella, M. Tessi

Synthesis of P-Stable Abstractions — SEFM 2020
A. Becchi, A. Cimatti, E. Zaffanella

Revisiting Polyhedral Analysis for Hybrid Systems — SAS 2019
A. Becchi, E. Zaffanella *Best Paper Award*

An Efficient Abstract Domain for Not Necessarily Closed Polyhedra — SAS 2018
A. Becchi, E. Zaffanella

A Direct Encoding for NNC Polyhedra — CAV 2018
A. Becchi, E. Zaffanella

THESIS

Abstractions for Reverse Engineering — Validating the Computerization of Relay-Based Railway Interlocking Systems — PhD 2025
A. Becchi

Software

PyCHC — Main developer

Open-source tool for certified Horn solving and design — 2025 – 2026

- Development skills: Python, PySMT library.
- Paper: *PyCHC: a Framework for Certified Horn Solving and CHC-based Design* (CAV 2026)

Golem — Contributor

Solver for Constrained Horn Clauses based on the OpenSMT solver — 2026

- Development skills: C++

ConvexQE — Main developer

Open-source personal project experimenting with Bidirectional Model-Based Projection — 2025

- Development skills: Python, PySMT library.
- Paper: *Syntactically Convex Model-Based Projection for Linear Real Arithmetic* (TACAS 2026)

NORMA — Contributor

Graphical modelling and analysis of relay-based railway interlocking systems — 2020 – 2025

- Developed the SMT-based optimizations and the automated property extraction.
- Development skills: Python, DIA frontend, nuXmv model checker, PySMT library and MathSAT solver.
- Papers: *NORMA: a Tool for the Analysis of Relay-Based Railway Interlocking Systems* (TACAS 2022); *Testing the Migration from Analog to Software-Based Railway Interlocking Systems* (CAV 2024); *Abstraction Modulo Stability* (FMSD 2024); *Abstraction Modulo Stability for Reverse Engineering* (CAV 2022)

LRA-BDD — Main developer

C++ library combining multi-terminal BDDs and convex polyhedra — 2020 – 2022

- Development skills: C++
- Papers: *Abstraction Modulo Stability* (FMSD 2024); *Abstraction Modulo Stability for Reverse Engineering* (CAV 2022); *Abstractions for Reverse Engineering* (PhD thesis 2025)

PPLite — Contributor

C++ library for the convex polyhedra abstract domain — 2017 – 2020

- Development skills: C++
- Papers: *PPLite: Zero-Overhead Encoding of NNC Polyhedra* (Inf. & Comp. 2020); *A Direct Encoding for NNC Polyhedra* (CAV 2018); *An Efficient Abstract Domain for Not Necessarily Closed Polyhedra* (SAS 2018)

PHAVerLite — Contributor

C++ verifier for hybrid systems — 2018 – 2019

- Development skills: C++
- Paper: *Revisiting Polyhedral Analysis for Hybrid Systems* (SAS 2019)

Teaching & Supervision

Teaching Assistant	*Computer Aided Verification*, USI	*2026/27*
	Theory of Computation, USI	*2026*
Thesis Co-supervisor	*Quantifier Elimination for Explainability* · Master's thesis	*2026/27*
	Inside the Golem's Head · Bachelor's thesis	*2026/27*

Academic Service

AE Co-Chair	VMCAI '27
Program Committee	NSAD '26
AE Committee	TACAS '25–'26 · SAS '22, '24 · CAV '22, '24, '26
Subreviewer	FASE '26 · SMT '26 · DATE '25 · FM '24 · CAV '20, '22, '24 · TACAS '20–'24, '27 · ATVA '21, '22, '26 · FMCAD '22, '26 · SEFM '20, '22 · SETTA '20
Student Volunteer	CAV '24 · FMCAD '22 · FLoC '18, '22

Seminars & Schools

Dagstuhl Seminars	*Software Contracts Meet System Contracts*	*Jan 2026*
	Sound Static Program Analysis in Modern Software Engineering	*Oct 2025*
	Theoretical Advances and Emerging Applications in Abstract Interpretation · collector	*Jul 2023*
Summer Schools	*Abstract Interpretation* · Lipari	*Sep 2024*
	Safety and Security through Formal Verification · Marktoberdorf	*Aug 2023*
	Verification Technology, Systems & Applications · EuroProofNet	*Sep 2022*

Languages

Italian	Native
English	B2, First Certificate in English (FCE)